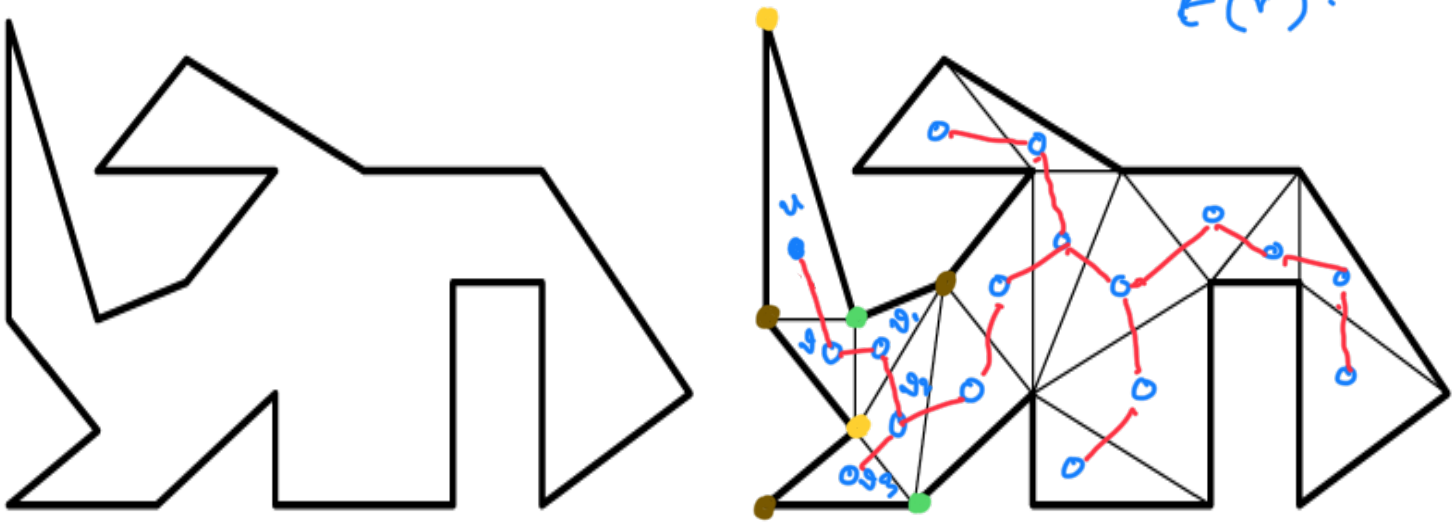
Art Gallery Problem – Triangulation

- Triangulation: A decomposition of a polygon into triangles by a maximal set of non-intersecting diagonals is called a triangulation of the polygon.

$t(u)$: triangle of u
$t(v)$: ... v



P T_P

Dual graph of T_P = $G(T_P)$

Art Gallery Problem

T_P : Triangulation of P

How do you find T_P, i.e. a triangulation of P.

- Dual graph of T_P

Write a pseudocode of 3-coloring of P with n vertices (T_P.)

Art Gallery Problem – Triangulation

Theorem 3.1 *Every simple polygon admits a triangulation, and any triangulation of a simple polygon with n vertices consists of exactly $n-2$ triangles.*

Art Gallery Problem – Coloring

Theorem 3.2 (Art Gallery Theorem) *For a simple polygon with n vertices, $\lfloor n/3 \rfloor$ cameras are occasionally necessary and always sufficient to have every point in the polygon visible from at least one of the cameras.*

$\lfloor n/3 \rfloor \rightarrow$ sufficient

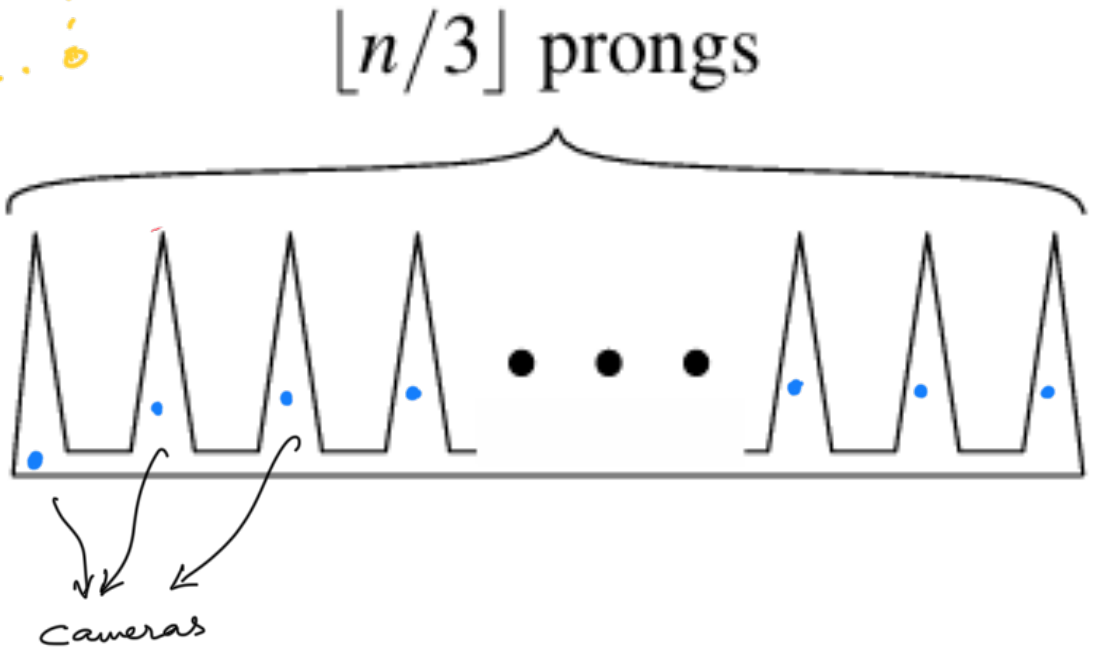
$a \rightarrow b$ a is sufficient for b

b is necessary for a

$\neg b \rightarrow \neg a$

Art Gallery Problem – Coloring

Occasionally necessary and always sufficient



Art Gallery Problem

- Suppose, we have P with n vertices. Also, a set of diagonals that partition P into convex quadrilaterals.
  How many cameras are sufficient?

- Does any polygon admit triangulation even if it has holes?
  Holes are also polygons.

